Fiscal Year:	FY 2020	**Task Last Updated:**	FY 05/29/2020
PI Name:	Chatterjee, Anushree Ph.D.		
Project Title:	Nucleic Acid Therapy Platform for Real-Time Countermeasures During Spaceflight Missions		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	TRISH--TRISH		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	80303-1904	**Congressional District:**	2
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2018 TRA BRASH1801: Translational Research Institute for Space Health (TRISH) Biomedical Research Advances for Space Health
Start Date:	01/01/2019	**End Date:**	09/30/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 9/30/2021 per E. Urquieta/TRISH (Ed., 5/17/21) NOTE: End date changed to 4/30/2022 per TRISH (Ed., 5/26/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Nagpal, Prashant Ph.D. (University of Colorado at Boulder) Zea, Luis Ph.D. (University of Colorado at Boulder)		
Grant/Contract No.:	NNX16AO69A-T0406		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	In long space exploration missions, astronauts are subjected to increased amounts of chronic radiation exposure, which can lead to Acute Radiation Syndrome (ARS) causing DNA damage, tissue damage, and aging. The goal of the project is to develop a high-throughput, customizable, gene-specific therapeutic strategy that will allow targeting of any gene for countermeasures, the design of an on-site development, and deployment of therapy for astronauts in need. Recent research efforts using nucleic acid therapy of mammalian cells include microRNAs, small interfering RNAs, long non-coding RNAs, deactivated CRISPR-Cas9 technology; however, though promising, these therapeutic strategies involve tedious cloning and optimization that are time-consuming, require specific skills that may not be available aboard International Space Station (ISS). The proposed research offers a drastic new approach with the development of a Facile Accelerated Specific Therapeutic (FAST) pipeline to design, build, and test treatment within a time scale of hours to days against any mammalian gene of interest by outlining the following tasks. (Aim 1) Facile and Accelerated high throughput design of peptide nucleic acids (PNAs) based antisense therapeutics that can block translation of any desired gene by binding to its mRNA for targeted inhibition. (Aim 2) Automated and high throughput parallel synthesis of PNA therapeutic library for generation of on-site therapeutic molecules, and development of nanoparticle as well as bacterial Type III secretion system based probiotic system to efficiently deliver therapeutic PNA molecules to host cells to overcome challenges of transport, toxicity, and bioavailability. (Aim 3) Testing of designed and built PNAs in a high throughput manner in a relevant macrophage, neutrophil, and hematopoietic stem cells based models, as well as a macrophage-infection model in presence of gamma-radiation. The proposed method will allow identification of important gene targets, and quickly generate translatable therapies that can be tested under host conditions, and deployed on-site within a time scale of hours to days. It is expected that the FAST pipeline can be deployed in the future in spaceflight settings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	EARTH AND SPACE: Development of platform technology to design countermeasure therapeutics rapidly to target almost any gene involved in radiation response. This can be valuable for cancer treatments. Demonstration of gene-specificity, reduced radiation response, increased transport, the lowered toxicity of PNAs. Novel nucleic acid/peptide purification and delivery strategy that can significantly impact broader fields of medicine with the capability to build disease and patient specific therapeutics. Development of novel therapeutics at an accelerated pace (< 1 week) that can be beneficial for broad range of countermeasures (such as new disease outbreaks). SPACE: Possibilities of deploying FAST platform on International Space Station (ISS) to provide radiation countermeasures (as well as other countermeasures--such as new antibiotics to treat drug-resistant infections). Automated high throughput PNA synthesis will reduce the cost of production as well as provide a method of PNA synthesis on-site (space station), therefore removing the need for long-term storage. The nanotechnology approach will provide a delivery vehicle for enhanced PNA transport, lowered toxicity, and increased bioavailability.
Task Progress:	In long space exploration missions, astronauts are subjected to increased amounts of radiation exposure, which can lead to Acute Radiation Syndrome (ARS) causing DNA damage, tissue damage, and aging. Immune dysfunction associated with spaceflight conditions can also potentially increase the susceptibility of crew members to pathogenic bacteria in extended space. We have developed a Facile Accelerated Specific Therapeutic (FAST) platform that can develop countermeasures and preventative agents for relieving ARS in less than a week with potential applications in future space missions. The FAST platform is high-throughput, and generates customizable, gene-specific therapeutics with high purity that targets genes relevant for radiation countermeasures in human cell lines. The FAST pipeline has accelerated design, build, test cycles, which allow drug discovery to testing to occur within a time scale of a week against any mammalian gene of interest by outlining the following steps: 1) Development of a bioinformatics toolbox for facile and accelerated high throughput design of peptide nucleic acids (PNAs) that act as inhibitors or activators of gene expression of specific targeted genes in less 3 hours. Therapeutic inhibition of expression is achieved by binding to mRNAs of targeted genes to block translation. Gene expression activation is achieved by building PNA activators that can bind to the respective promoter regions and recruit transcriptional activators. 2) Automated and high throughput parallel synthesis of PNA inhibitor/activator library for generation of on-site therapeutic molecules is achieved within 3-4 days. Further, we have developed a nanoparticle based transport system for high-throughput bulk purification, as well as efficient delivery of therapeutic PNAs to host cells at sub-micro molar range to overcome challenges of transport, toxicity, and bioavailability. 3) Testing of designed and built PNAs in a high throughput manner in relevant human cell lines to demonstrate their efficacy in regulating expression of radiation-relevant genes as desired. FAST allows for identification of important gene targets, and quickly generates translatable therapies that can be tested under host conditions. The accelerated design, build, and test cycles of FAST platform allow for effective countermeasures in less than a week, as well as development of a prototype that could be potentially deployed in future flight mission to serve medical needs for astronauts. The FAST platform can have significant impact for applications both on space and Earth.
Bibliography Type:	Description: (Last Updated: 10/31/2025)
Articles in Peer-reviewed Journals	Levy M, Bertram JR, Eller KA, Chatterjee A, Nagpal P. "Near-infrared-light-triggered antimicrobial indium phosphide quantum dots." Angew Chem Int Ed Engl. 2019 Aug 12;58(33):11414-8. Epub 2019 Jul 4. https://doi.org/10.1002/anie.201906501 ; PubMed PMID: 31184802 , Aug-2019
Awards	Chatterjee A. "Chair's Faculty Fellowship, June 2019." Jun-2019
Awards	Chatterjee A. "Provost Faculty Achievement Award, October 2019." Oct-2019
Awards	Chatterjee A. "VastLife Sciences Award for studying human microbiome, June 2019." Jun-2019
Awards	Nagpal P. "Department of Energy Award, September 2019." Sep-2019